

Assignment #4 Part (II)

Course: ISA 345

Instructor: Dr. Maliha Alam

Points: 70

Submission instructions: this assignment is to be done individually. All your answers must be in a single document, *e.g.*, a *doc* or a *pdf* file. For database-modeling questions, you are allowed to take print screens of your models and insert the resulting figures into your document. Another option is to export diagrams to figures by clicking at “File -> Data Modeler -> Print Diagram -> To Image File” in Oracle SQL Developer, and then import these figures into a text editor such as MS Word. Data-modeling questions must be answered using the Data Modeler tool inside Oracle SQL Developer. Reverse engineering is allowed (*i.e.*, creating diagrams from DDL). Unreadable models will receive zero points Submit the final document on Canvas before the due date.

Question 1: Assume you are working for an IT company hired to develop a database for Walgreens, the famous chain of pharmacies. The information below is a summary of the requirements document created during initial meetings with key stakeholders.

- a) Draw a logical database model that captures the following information. In your logical model, all relationships must be named on both ends. Cardinalities and modalities must also be defined. **[15 points]**

Clearly, patients are at the core of Walgreens’ business. Currently, patients are uniquely identified by their insurance number. It is also a requirement that each patient’s name, address, and date of birth must be recorded.

Every patient has one primary physician, whereas every physician has at least one patient. Physicians are uniquely identified by their license number. Each physician’s name and specialty must also be recorded.

A physician (respectively, patient) can write (respectively, obtain) one or more prescriptions. Each prescription contains data that identify the physician who prescribed it, the patient who received the prescription, a unique prescription number and the date of the prescription.

Besides the above information, a prescription contains one or more drugs. Clearly, a drug can be part of many prescriptions, and new drugs might not have been prescribed yet. The unique trade name and formula of each drug must be recorded. One drug may be derived from another single drug, whereas a single drug may be the source of many other drugs.

Each drug is sold to Walgreens by a given pharmaceutical company. Each pharmaceutical company in the database is identified by its name, has a phone number, and sells at least one drug to Walgreens.

- b) Translate the model in a) into a *physical database model*. In your physical model, all attribute types must be defined, and the relationships must be sound. **[10 points]**

Question 2: During a consulting job for a company working on green-energy products, you come across an in-house project management system that relies on a badly designed database. Being a good-hearted person, you decide to fix the database by performing normalization. A sample of the data stored in the current database is displayed in the following page. The database consists of a single table containing six attributes related to projects and employees. The database contains many candidate keys, but the composite primary key is predefined as (*ProjNo*, *EmpNo*).

- a) Redesign the original database so that it satisfies the second normal form. You do not have to list FDs for all candidate keys (there are many of them!!!). Instead, list all the FDs where each individual component of the PK as well as the composite PK are the determinants. Use your FD list to create new tables (see Lectures 20 and 21). Give suitable names to the new tables and show how the sample data in Table 1 are now stored in the new tables. There is no need to code the tables in SQL. **[15 points]**
- b) Convert the new design from 3 - a) to the third normal form. Remember that you have to check whether or not each new table from 3 – a) is in 3NF. If a table is not in 3NF, then you have to split that table so that the 3NF requirements are satisfied (see Lecture 22). In particular, for each table from 3 – a), start by listing the transitive dependencies $X \rightarrow Y \rightarrow Z$, where

- X is a candidate key and Y is a non-candidate key. Similar to what you did in 3 – a), focus on cases where X is either *ProjNo*, *EmpNo*, or (*ProjNo*, *EmpNo*). Give suitable names to the new tables and show how the sample data in Table 1 are now stored in the new tables. There is no need to code the tables in SQL. **[10 points]**
- c) Based on the new database design from 3 – b), create a physical database model using Data Modeler where attribute types and the relationships between attributes are clearly specified (*i.e.*, the PK-FK pairs). **[10 points]**
- d) Reflect on the new database model. In particular, explicitly describe by means of examples 3 anomalies (one insertion, one deletion, and one update anomaly) that were present in the original design of the database and that are no longer present in the final design you developed in 3 – c). **[10 points]**

Table 1: Sample from the original database

ProjNo	ProjName	EmpNo	JobType	PriceHour	WorkHours
15	Solar City	103	Engineer	84.50	23.8
15	Solar City	101	Database Designer	105	19.4
15	Solar City	105	Database Designer	105	35.7
15	Solar City	106	Programmer	35.75	12.6
15	Solar City	102	Systems Analyst	96.75	23.8
18	Fuel Cell	114	App Designer	48.10	24.6
18	Fuel Cell	118	General Support	18.36	45.3
18	Fuel Cell	104	Systems Analyst	96.75	32.4
18	Fuel Cell	112	DSS Analyst	45.95	44
22	Chargers V3	105	Database Designer	105	64.7
22	Chargers V3	104	Systems Analyst	96.75	48.4
22	Chargers V3	113	App Designer	48.10	23.6
22	Chargers V3	111	Clerical Support	18.36	22
22	Chargers V3	106	Programmer	35.75	12.8
25	Hybrid EV	107	Programmer	35.75	24.6
25	Hybrid EV	115	Systems Analyst	96.75	45.8
25	Hybrid EV	101	Database Designer	105	56.3
25	Hybrid EV	114	App Designer	48.10	33.1
25	Hybrid EV	108	Systems Analyst	96.75	23.6
25	Hybrid EV	118	General Support	18.36	30.5
25	Hybrid EV	112	DSS Analyst	45.95	41.4